Projectile, Problem — Solution.

(1) horizontal range = maximum height

$$\frac{u^2 \sin 2\theta}{g} = \frac{u^2 \sin^2\theta}{2g}$$

$$2 \sin\theta \cos\theta = \sin\theta \times \sin\theta$$

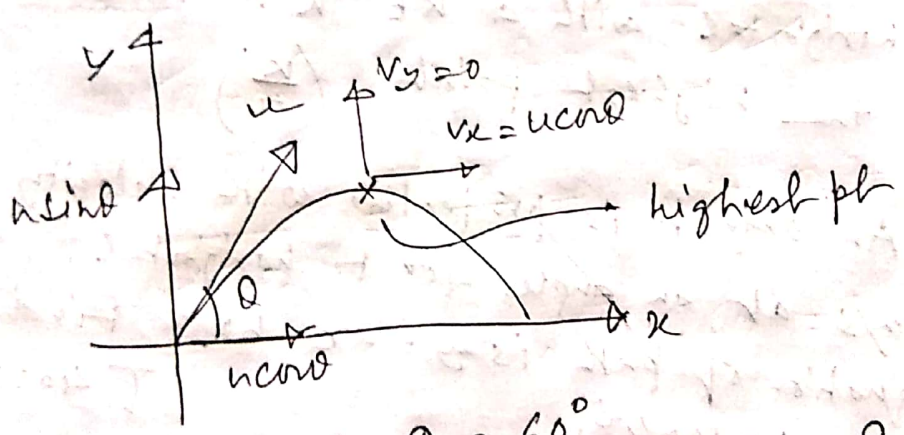
$$\tan\theta = 4 \quad \therefore \boxed{\theta = \tan^{-1}(4)}$$

(2) We know that, for two angle of projections θ and $(90^\circ - \theta)$, range (R) is same. So here

Range for angle 30° say R_1 = Range for angle 60° say R_2

$$\therefore R_1 = R_2 \quad \text{or} \quad \boxed{\frac{R_1}{R_2} = 1:1}$$

(3)



$u = 5\ m/s$ & $\theta = 60^\circ$,

velocity at highest point $= u\cos\theta$
$= 5\cos 60^\circ$
$= 5/2\ m/s$

So linear momentum (রৈখিক ভরবেগ) at highest point $= mv_x = mu\cos\theta$
$= 4 \times \frac{5}{2} = 10\ kg\,m/s$

Scanned with CamScanner

(4) Range of a projectile for angle of projection θ = $\frac{u^2 \sin 2\theta}{g}$

for max^m range, $\theta = 45°$ $\therefore$ max^m range = $\frac{u^2}{g}$

So acc to problem, $R = \frac{u^2}{g}$

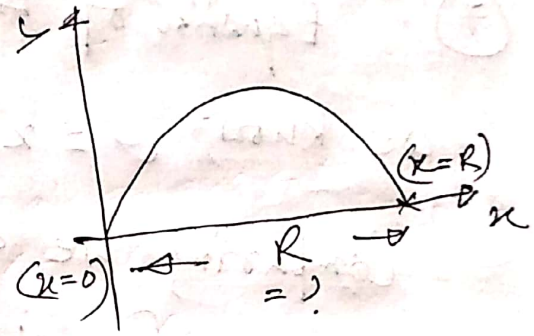
Hence Corresponding max^m height

$$H = \frac{u^2 \sin^2\theta}{2g}$$

$$= \frac{u^2 \sin^2(45°)}{2g}$$

$$= \frac{u^2}{g} \times \frac{1}{4} = \frac{R}{4}$$

(5) eqn of projectile is



$$y = 12x - \frac{3}{4}x^2$$

put $y = 0$,

$$0 = 12x - \frac{3}{4}x^2$$

$$48x - 3x^2 = 0$$

$$x(48 - 3x) = 0 \quad \therefore x = 0 \text{ [starting point]}$$

or $48 - 3x = 0$

$\therefore$ $x = 16m$

$\therefore$ Range = 16 m

[Note to find range from eqn of projectile, put $y = 0$, then non zero value of x will give you range]

Scanned with CamScanner

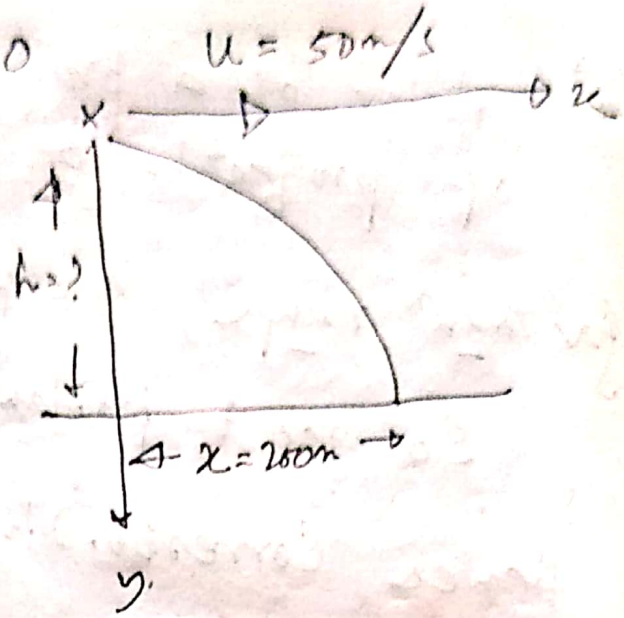
(6) $\because$ x is known $= 200m$

$\therefore$ along x axis $\rightarrow$

$\boxed{x = ut}$ — (1)

$200 = 50t$

$t = 4 sec$



& along y axis

$\boxed{y = \frac{1}{2} g t^2}$ — (2) ($\because$ along y, $u = 0$)

$h = \frac{1}{2} \times 10 \times (4)^2 = 80m$

(7) Equation of path $y = \sqrt{3}x - \frac{g}{2}x^2$

we know, $y = x \tan\theta - \frac{g}{2u^2\cos^2\theta} x^2$

Comparing we get

$\tan\theta = \sqrt{3}$ $\therefore$ $\boxed{\theta = 60^\circ}$

& $\frac{\cancel{g}}{\cancel{2}u^2\cos^2\theta} = \frac{\cancel{g}}{\cancel{2}}$ $\Rightarrow$ $u^2\cos^2\theta = 1$

or, $u^2 \cos^2 60^\circ = 1$

or, $u^2 \times \frac{1}{4} = 1$

$u^2 = 4$ or $\boxed{u = 2m/s}$

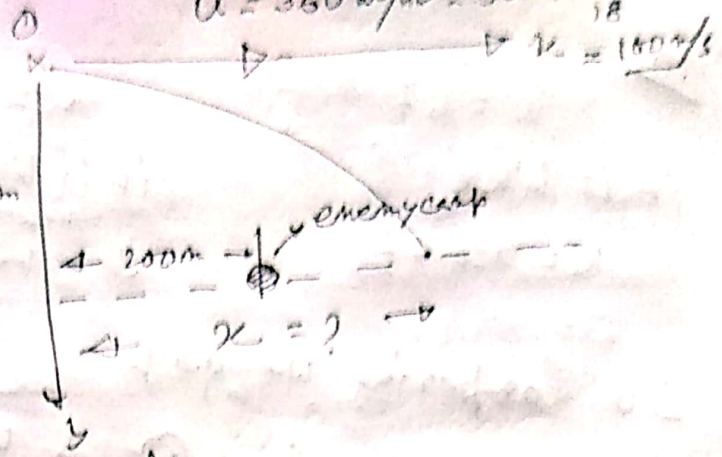
Scanned with CamScanner

(8) Note: $km/hr \times \frac{5}{18} \Rightarrow m/s$

$u = 360\ km/hr = 360 \times \frac{5}{18}$

$u = 100\ m/s$

$u = 360\ km/hr = 360 \times \frac{5}{18} = 100\ m/s$



$\because$ y is known (500m)

$\therefore y = \frac{1}{2} g t^2$

$500 = \frac{1}{2} \times 10 \times t^2$

$t^2 = 100 \Rightarrow \boxed{t = 10\ sec}$

$\therefore x = ut = 100 \times 10$

$= 1000\ m$.

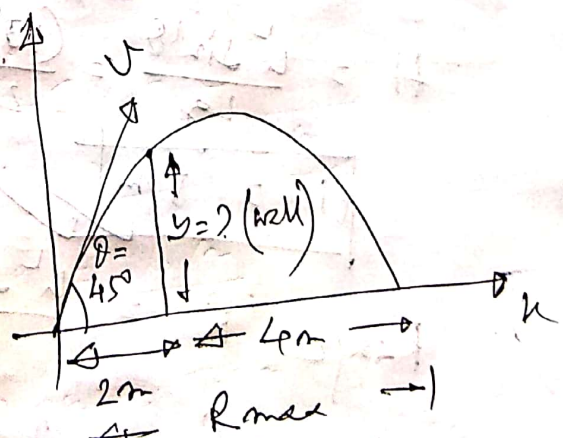
So the bomb will land at a horizontal distance 1000 m. Since the enemy camp is at horizontal distance 200 m, so the bomb will not hit the enemy camp, it will fall ahead of it by 800 m.

(9) $\because \theta = 45°$

$\therefore R = R_{max}$.

$R_{max} = \frac{u^2}{g} = 2 + 4 = 6m$

$\boxed{\frac{u^2}{g} = 6}$.



$\therefore$ ht of wall

$y = x \tan\theta - \frac{g x^2}{2u^2 \cos^2\theta}$

$y = 2 \tan 45° - \frac{1 \times (2)^2}{2 \times 6 \times \cos^2(45°)} \Rightarrow y = 2 - \frac{4 \times 2}{2 \times 6 \times 3}$

$\boxed{y = \frac{4}{3} m}$

Scanned with CamScanner

10. $x = 6t$, $y = 8t - 5t^2$

it is like $x = u\cos\theta\, t$, it is like $y = u\sin\theta\, t - \frac{1}{2}gt^2$

So Comparing we get

$u\cos\theta = 6$ — (1) & $u\sin\theta = 8$ — (2)

(1)^2 + (2)^2, $u^2\cos^2\theta + u^2\sin^2\theta = 6^2 + 8^2$

$u^2 = 100$

$\boxed{u = 10\,m/s}$

for (2) $u\sin\theta = 8 \Rightarrow 10\sin\theta = 8$

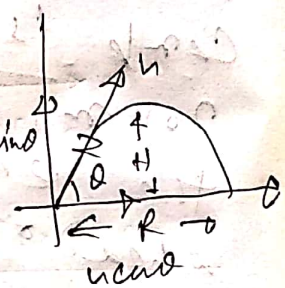
$\sin\theta = \frac{4}{5}$

or $\boxed{\theta = \sin^{-1}(4/5)}$

(11) $\vec{u} = 3\hat{i} + 10\hat{j}$

like, $\vec{u} = u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}$

$\therefore \boxed{u\cos\theta = 3}$ $\boxed{u\sin\theta = 10}$



$H = \frac{u^2\sin^2\theta}{2g} = \frac{(10)^2}{2\times10} = \frac{10}{2}$

$\boxed{H = 5\,m}$

Range $R = \frac{u^2\sin 2\theta}{g} = \frac{u^2 \times 2\sin\theta\cos\theta}{g}$

or, $R = 2\times\frac{(u\sin\theta)\times(u\cos\theta)}{g}$

$= 2\times\frac{3\times10}{10} = 6\,m$

$\boxed{R = 6\,m}$

Scanned with CamScanner

(12) $\vec{u} = \hat{i} + 2\hat{j}$

It is similar to, $\vec{u} = u_x\hat{i} + u_y\hat{j}$

$\therefore$ u_x (horizontal component of u) $= 1$

& u_y (vertical component of u) $= 2$

Now after time t along x axis

$x = u_x t \Rightarrow x = 1 \times t$ — (1)

& $x = t$

& along y axis,

$y = u_y t - \frac{1}{2} g t^2$

$y = 2t - \frac{10}{2} t^2$

$y = 2t - 5t^2$ — (2)

put $t = x$ from (1)

$\therefore$ eqn of path between y & x is

$y = 2x - 5x^2$

(13) for a projectile thrown with velocity 'u' at angle of projection 'θ', eqn of path is $y = x\tan\theta - \frac{g}{2u^2\cos^2\theta}x^2$

So, $y = x\tan\theta - \frac{x^2}{2\frac{u^2\cos^2\theta}{g}}$

We know $R = \frac{2u^2\sin\theta\cos\theta}{g}$

or, $y = x\tan\theta - \frac{x^2}{\left(\frac{2u^2\sin^2\theta\cos^2\theta}{g}\right)\left(\frac{\cos\theta}{\sin\theta}\right)}$

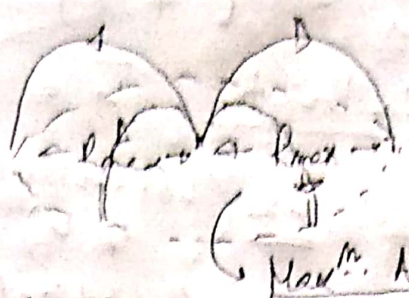
OR $y = x\tan\theta - \frac{x^2}{R}\tan\theta$ or $y = x\tan\theta\left(1 - \frac{x}{R}\right)$

Scanned with CamScanner

(14)

Range of bullet

$R = \frac{u^2 \sin 2\theta}{g}$



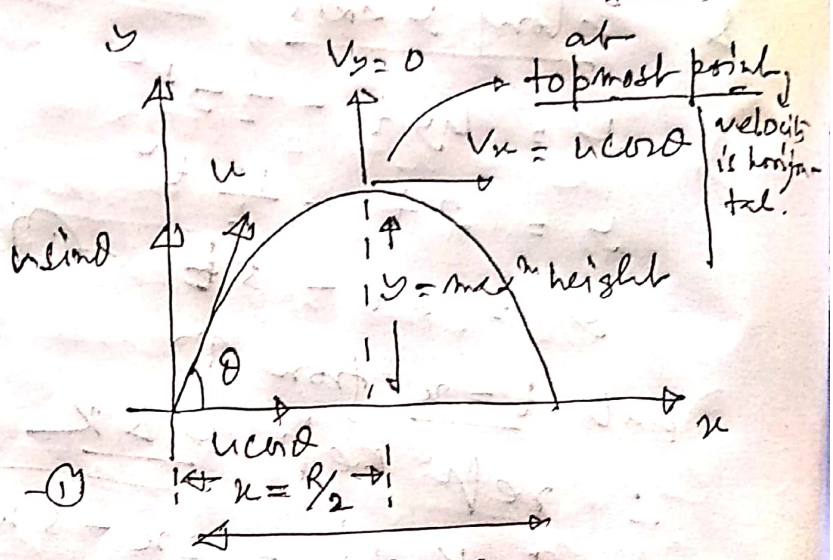
for max^m range

$\sin 2\theta = 1$

$\therefore R_{max} = \frac{u^2}{g}$. This is the radius of the desired circle. So max^m area of the circle $A_{max} = \pi R_{max}^2$

$$= \pi \left(\frac{u^2}{g}\right)^2 = \frac{\pi u^4}{g^2}$$

15.



Here

$x = \frac{R}{2}$

$= \frac{2u^2 \sin\theta\cos\theta}{2 \times g}$

$x = \frac{u^2 \sin\theta\cos\theta}{g}$ — (1)

$y = \frac{u^2 \sin^2\theta}{2g}$ — (2)

$\boxed{u^2 \sin^2\theta = 2gy}$ — (A) & $\frac{x^2}{2y} = \frac{\cancel{u^4} \sin^2\theta \cos^2\theta}{\cancel{g^2} \times \cancel{2} \times \frac{\cancel{u^2} \sin^2\theta}{\cancel{2g}}}$

or, $\frac{x^2}{2y} = \frac{u^2 \cos^2\theta}{g}$

So $\boxed{u^2 \cos^2\theta = g\frac{x^2}{2y}}$ — (B)

Now, (A)+(B), $u^2 \sin^2\theta + u^2 \cos^2\theta = 2gy + g\frac{x^2}{2y}$

or, $u^2 = g\left(2y + \frac{x^2}{2y}\right)$

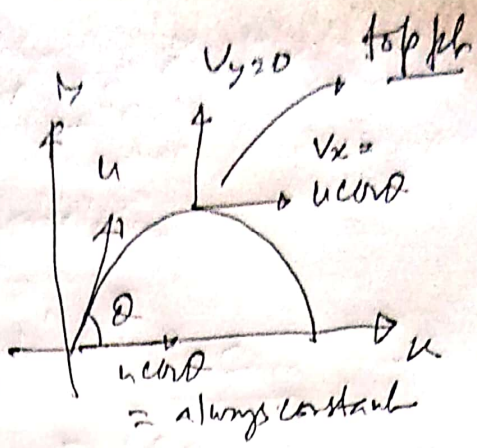
$\therefore u = \left\{g\left[2y + \frac{x^2}{2y}\right]\right\}^{1/2}$

Scanned with CamScanner

16/ (done in class),

KE at the point of projection is

$\frac{1}{2}mu^2 = K$ (given)



KE at max^m height

$= \frac{1}{2}m(u\cos\theta)^2$

$= \left(\frac{1}{2}mu^2\right)\cos^2\theta = K\cos^2\theta$

$= K\cos^2 60^\circ$

$= \frac{K}{4}$

(done in class)

(17) given $\boxed{y = ax - bx^2}$ — (1)

we know, $y = x\tan\theta - \frac{g}{2u^2\cos^2\theta}x^2$

$\because \theta = 45^\circ$, & $u = 20\,m/s$,

$\therefore y = x - \frac{10}{2\times(20)^2\times\cos^2 45^\circ}x^2$

$\boxed{y = x - \frac{1}{40}x^2}$ — (2)

Compare (1) & (2), $a = 1$, $b = \frac{1}{40}$

$\therefore a : b = 1 : \frac{1}{40} = 40:1$

(18) (done in class)

given $u_x = 4\,m/s$ & $u_y = 0\,m/s$

velocity after time $t = 0.3$ sec

$\vec{V} = V_x\hat{i} + V_y\hat{j} = (u_x + a_x t)\hat{i} + (u_y + a_y t)\hat{j}$

$[\because a_x = 0,\ a_y = +g = +10\,m/s^2]$

$\vec{V} = (4)\hat{i} + (0 + 10\times 0.3)\hat{j}$

$\boxed{\vec{V} = 4\hat{i} + 3\hat{j}}$ $\therefore |\vec{V}| = \sqrt{4^2 + 3^2} = 5\,m/s$

& angle made by $\vec{V}$ with ground i.e. x axis is $\tan\theta = \frac{3}{4}$ or, $\boxed{\theta = \tan^{-1}\left(\frac{3}{4}\right)}$

Scanned with CamScanner